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THE CURRY QUEST

robots meet ARTS

Subject:
Mother language

Content:
Understanding and expressing oneself orally. Producing written works. Understanding how language works. Literary and artistic culture.

Computational practices: choose what fits

- Decomposing
- Pattern recognition
- Abstraction
- Algorithms
- Problem solving

Unplugged/plugged:
Unplugged

Preparation time:
45 min

Age:
8-12 years

Duration:
4 hours

Location:
In class

Group distribution:
In groups of 3-5 students

Difficulty

"Understanding and expressing oneself orally", "Writing", "Understanding how language works", "Literary and artistic culture"
"The Curry Quest" develops language skills by having students collaboratively create an interactive, branching narrative. They practice oral expression through the exchange of ideas, master writing by developing various narrative branches, and learn the characteristics of a specific literary genre.

"Sensitivity", "Judgment"
This activity stimulates students' sensitivity, requiring them to listen and empathize to integrate each other's ideas, while developing their critical thinking to evaluate different narrative choices and their consequences.

"Search", "Reason", "Communicate"
"The Quest for Curry" mobilizes mathematical skills by leading students to logically organize their story, to solve problems of interconnection between narrative branches, and to use visual representations (flowcharts) to model their story, establishing a concrete link between narration and logical reasoning.

“Practicing scientific and technological approaches”, “Designing”, “Appropriating tools and methods”
The activity integrates scientific practices by engaging students in a design process where they model their story, test the coherence of narrative paths, and potentially implement their creation in a digital environment, allowing them to acquire systematic working methods common to science.

Materials needed:

- Large sheets of paper or whiteboards for modeling
- Markers of different colors
- Dice (6-sided or special)
- Card game or special cards to create
- Blank flowchart templates
- Explanatory sheets on conditional structures
- Simple examples of narrative mazes

Helpful links:

- Digital tools and platforms
  - Scratch (https://scratch.mit.edu)
  - Genially (https://genial.ly)
  - MakeCode Arcade (https://arcade.makecode.com)
  - Twine (https://twinery.org)
  - Book Creator (https://bookcreator.com)
  - Kamishibaï (https://kamishibais.com/)
- Narrative structures
  - Hero’s jounrey (https://en.wikipedia.org/wiki/Hero%27s_journey)
  - Propp's story cards
  - Actancial model (https://en.wikipedia.org/wiki/Actantial_model)
- Algorigrams and computational thinking
  - Flowcharts (https://en.wikipedia.org/wiki/Flowchart)
  - Mermaid (https://mermaid.live)
- Examples of game books and interactive stories
  - The “Défis Fantastiques” library (https://www.gallimard.fr/)
  - “Choose your own adventure” books (https://en.wikipedia.org)

Introduction game "What if…"

Seat the students **in a circle** in an open area of the classroom. Start the activity by proposing a **simple and evocative starting situation**: "You enter a mysterious cave..."

A first student receives the die, throws it, and depending on the result, continues the story according to the following rules:

- If the number obtained is **even**, the student must imagine a **positive event** ("You discover a chest full of treasures")
- If the number is **odd**, the student must introduce an **obstacle or difficulty** ("A landslide suddenly blocks the entrance to the cave").

The die is passed from student to student, each adding an element to the collective story while respecting the odd/even rule, thus **creating a collaborative narrative with unpredictable twists and turns.**

This fun activity allows you to concretely introduce the notion of **conditional choice** while stimulating narrative imagination.

Discovery through experience

Present your class with a **short example of a narrative maze** that you have prepared in advance. This interactive mini-story, which you could title "**Choosing the Path**," should only have **2 to 3 simple but illustrative branches.**

For example, the protagonist comes to a fork in the forest and must choose between taking the lighted path or the shaded passage. **Read the story aloud** and, at each decision point, **invite students to collectively determine which path to follow using the mechanism you have planned** (rolling dice, drawing cards).

Ask students to **note in their notebooks the specific moments when a choice arises and how this choice influences the rest of the story.**

Once the example is complete, lead a brief discussion to **explore the different mechanisms that can determine the outcome of a choice**: **chance** (dice, cards), **conditional logic** (if… then…), or **skills assigned to the character**.

This hands-on experience will allow your students to **visualize the structure** of an interactive story before embarking on creating their own.

Preparation and teacher notes

- Prepare some short examples of stories with simple branches
- Provide one die per group
- Prepare game cards with different symbols or keywords
- Arrange the tables in islands to facilitate work in groups of 3-5 students

Warm up

<u>Introduce the sequence</u>

The activity "**The Curry Quest - A Choose Your Own Adventure**" invites students to create an interactive story, inspired by "**choose your own adventure**" books, where the reader (or player) must make choices at each stage to guide the narrative.

Specifically, students, grouped into small groups, begin by defining a narrative framework around a chosen theme (for example, a historical adventure or a fantasy quest). They then identify key decision points in the story, where several options are available to the reader.

At each branch, a **condition is associated with the proposed choice**. This condition can be determined by various mechanisms—**such as rolling a die, using a card, or any other simple rule**—that are reminiscent of conditional structures in algorithms (**such as "if... then..."**). For example, "**if the die shows an even number, the hero crosses the forest; otherwise, he bypasses the village.**" These conditions introduce an element of randomness and make the story interactive and unpredictable.

A key aspect of this activity is **algorithmic modeling**. Once the story is designed, each group of students translates all the branches into a **flowchart**. This visual representation helps verify the narrative's coherence: it identifies whether **all branches lead to a satisfying conclusion** or whether some **lead to dead ends**. Thus, modeling serves both to illustrate a pre-existing narrative and to guide the construction of a story that always ends happily.

Discussion on interactive games and books

Start a conversation with your class about the **interactive narratives that already exist in our cultural environment**. Mention popular "**choose your own adventure**" collections, as well as **narrative video games** where players directly influence the story (such as some adventure games or RPGs). Also discuss **tabletop role-playing games**, which operate on the same principle of interactive storytelling.

Then invite your students to share **their own experiences**: “Who has ever read a choose-your-own-adventure book? Who has ever played a video game where your choices change the story?” Encourage them to describe how they felt when they made those choices and how it changed the way they followed the story. **Write down any relevant points they mention on the board.**

Use these testimonies to introduce, without using technical terminology, the **algorithmic concepts** that underlie these stories: **"In these stories, notice how a choice sends you to a specific page... It's as if the author had planned a particular path for each possible decision."**

Finish by explaining that the activity they will carry out following this introduction will allow them **to understand how these stories are constructed, and even to create some themselves.**

Reflection on the warm up sequence

Gather your students for a moment of sharing and reflection on the **experience** they have just had. Lead an informal discussion focused on the **emotions** and **reactions** provoked by the activities: “**How did you feel** when you had to make a choice in the story?”, “**Which did you prefer**: deciding what happens next or discovering the consequences of your choices?”, “**Were you surprised** by any of the outcomes?” This conversation will allow students to **become aware** of the emotional dimension of interactive stories and the **particular engagement** they generate in the reader or player.

Then announce the **goal for the next session**: to work in small groups to **create their own**, more elaborate narrative maze, with multiple branches and choices with varying consequences. Mention that they will also learn how to **visually represent their story** structure to ensure that all paths lead to a satisfying ending.

In the appendix you will find **several examples of short branching stories** suitable for this sequence, which you can use as models or as sources of inspiration.

Preparation and teacher notes

Prepare blank A3 **flowchart templates** for each group, along with a few more elaborate examples to project.

Gather **simple materials on classic narrative structures** (such as the hero's journey diagram) adapted to the students' level.

Create **illustrated explanatory sheets** on **conditional structures, with concrete examples**. Also provide markers of different colors, large sheets of paper, and sufficient display space for the productions.

Build up

Introduction and educational objectives

Clearly present the objectives of this activity: **to deepen the concept of interactive storytelling by creating a more complex story and learning how to model it visually**. The idea is to show how, at each stage of the story, conditioned choices (for example, based on a die roll or a card choice) can influence the rest of the story.

Explain the **link with computational thinking** by simply introducing the algorithmic concepts they will discover: **conditions, branches and loops.**

Highlight the skills that this activity will help develop in your students:

- **logic** through the development of coherent conditional structures,
- **narrative creativity** through the invention of stories with multiple possibilities,
- **modeling** through the visual representation of complex processes.

Narrative design phase

Divide the students into **groups of 3 to 5**. Invite each group to **choose a stimulating narrative theme adapted to their level** (fantasy adventure, detective investigation, time travel, space exploration, etc.).

Have them develop a **central narrative using a simple visual aid, such as a timeline or tree diagram.** Ask them to identify key moments where the reader will have to make an important choice.

For each of these decision points, students will need to clearly **define which mechanisms (dice, cards, other simple rules) will determine the direction of the story**. Circulate among the groups to help them structure their thinking and define a coherent narrative arc.

A guide sheet is available in the appendix to explore the **narrative structure of the monomyth.**

Construction of conditional branches

Distribute materials to each group to develop their interactive story: **paper, pencils, structured worksheets.**

Guide them through the creation of **at least three major decision points in their story.** For each branching story, guide them to **associate specific and explicit conditions with each choice, using the "if... then..." formula (e.g., "If the die rolls a 1 or 2, then the hero discovers a secret passage").** These rules introduce an element of chance or resource management, making the story interactive.

Emphasize the importance of **checking the narrative coherence of each branch** and ensuring that all lead to a satisfying conclusion. Encourage creativity and imagination.

Algorithmic modeling

Introduce **visual modeling tools** by presenting a simple example of a **flowchart.** Explain how to represent the different elements: **rectangles for actions or events, diamonds for decisions, arrows for transitions.**

Then, give each group a **large sheet of paper and markers** so they can **translate their story into a flowchart.** Guide them step by step through this representation, helping them **use the symbols correctly.** This visualization will allow them to identify any inconsistencies in their narrative structure and ensure that all paths lead to a satisfactory conclusion.

Introducing Narrative Mazes

Organize a **sharing of narrative mazes between groups**. Each group tests another group's narrative path by following the defined rules and **documenting the different paths taken.**

Encourage students to note their observations: **moments when they were surprised, passages that seemed confusing, endings that seemed satisfying or not.**

After these discussions, bring the students together for a **kind and constructive critical review of the flow and coherence of the mazes.** The groups can then make changes to their work based on the feedback received.

Reflexion on the Build up sequence

Conclude the session with a group discussion where **each group presents their final work.** Focus the discussion on the algorithmic concepts discovered through this activity: **the notion of condition, the structure of branching, and how these elements helped create a coherent narrative despite its complexity.**

Encourage them to **share the challenges they encountered** and the solutions they found, thus valuing their problem-solving process.

You can create **links between this activity and other disciplines**, such as **mother language** (narrative structure) or **mathematics** (logic).

To go further, you can present students with possibilities for extending their work for the next session: **transforming their narrative labyrinth into digital format, creating a small book, or developing a board game based on their story.**

Distribute a summary **sheet on the algorithmic principles discussed**, which they can keep as a reference. Suggest that students who are particularly interested **research examples of video games or apps** that use these same interactive narrative principles.

Preparation and teacher notes

Prepare **concrete examples of digital implementation of interactive stories, such as Scratch or MakeCode Arcade projects** that you can show to students.

Gather a variety of resources on **creating educational escape games that are age-appropriate for your students.**

Prepare **simplified and illustrated tutorials for block programming**, with specific examples of conditional structures.

Make sure you have the necessary equipment for each type of possible extension: **computers or tablets** for digital options, **creative equipment** for unplugged options.

Rehearsal

Exploration phase: what adventure are we going on?

Introduce students to the **different possibilities for extending and fleshing out their narrative maze.** Show concrete examples of each format to inspire their creativity:

- A simple interactive web application made with **Genially**
- A mini-video game created with **MakeCode Arcade**
- A **kamishibai** (Japanese paper theater) for a visual and oral presentation
- An educational **escape game** with story-related puzzles
- A **board game** with cards, a die and a path corresponding to the story

Briefly present the advantages and constraints of each format.

Then, let the students reorganize into groups **according to their interests and affinities** for these different formats.

Creation phase: bringing our story to life

Accompany each group according to the option chosen, circulating from one group to another and providing the specific resources needed:

For groups working on MakeCode Arcade or Scratch	• Guide them through the discovery of conditional programming blocks • Show them how to create simple characters (sprites) and backgrounds • Help them implement narrative choices through basic interactions • Encourage them to simplify their story if necessary to make it programmable
For groups creating an escape game	• Help them turn their story into logic puzzles • Guide them in creating physical supports (clues, objects to manipulate) • Help them organize their escape game in space and time
For groups developing a board game	• Provide them with materials to create their game board with different paths • Help them design event and decision maps • Guide them in developing clear rules that incorporate conditional mechanisms
For groups working on kamishibai or Genially	• Show them how to adapt their storytelling to the chosen visual format • Help them create effective illustrations • Guide them in organizing the sequences to preserve interactivity

Testing phase: let's play and improve

Invite students to **develop a minimal but functional version** of their creation. Remind them that this is a **prototype**, not a final version.

Encourage them to focus on the essential aspects: **playability for games, clarity for visuals, functionality for digital versions**. Then organize an interactive presentation of the creations between groups, where each team tries out another's creation and provides constructive feedback. Allow time for the groups to make adjustments and improvements.

Reflexion on the Build up sequence

Organize a session where **each group presents their final creation to the entire class.** Provide a suitable space where the demonstrations can be easily seen by everyone.

After each presentation, lead a short discussion session where other students can **ask questions or make comments**. Make sure to **recognize each group's** work by highlighting strengths and original ideas. This step allows students to become aware of the diversity of approaches possible based on a single basic concept.

Lead a group discussion to summarize the algorithmic concepts learned during the three sessions: **conditional structures, modularity, variables, modeling, and problem decomposition**. Highlight the connections between storytelling and programming, **showing how narrative choices correspond to logical structures**. Encourage students to self-identify the cross-curricular skills they have developed: **creativity, logic, collaboration, critical analysis, and visual communication.** Write these elements on the board to create a mind map of learning.

Present to the students a vision of the **possible extensions** of this activity through the formalization of the progressive path they have followed and the next steps:

- **What they did**: created narrative labyrinths and transposed them into different formats
- **Possible next steps**: more advanced digital implementation, creation of more complex games
- **Future horizons**: projects integrating several disciplines, sharing with other classes

End with a quick roundtable discussion where each student shares one thing they particularly enjoyed about this sequence of activities.

Example of story Sequence 1

The Dangers of the Minotaur

Initial situation: The hero arrives in front of three doors.

- **If the die shows 1 or 3,** he chooses the left door and meets the Minotaur.
    - **If he rolls the die and gets an odd number,** he fights the Minotaur.
    - **If he rolls the die and gets an even number,** he escapes.

- **If the die shows 2 or 4,** he chooses the middle door and discovers a treasure.
    - **If he rolls the die and gets an even number,** he takes the treasure and triggers a trap.
    - **If he rolls the die and gets an odd number,** he leaves the treasure and continues on his way.

- **If the die shows 5-6,** he chooses the door on the right and encounters a puzzle.
    - **If he answers correctly,** he accesses the exit.
    - **Otherwise,** he must return to the beginning of the maze.

This simple structure illustrates the concept of conditional branching. The use of the Minotaur theme allows for a connection to mythology and a famous labyrinth.

Example of story Sequence 1

The Death Star Infiltration

Initial situation: A young Rebel infiltrates the Death Star and arrives in front of three corridors.

- **If the die shows 1 or 3**, he takes the left corridor and comes face to face with a group of Stormtroopers.
    - **If he rolls the die and gets an odd number**, he draws his blaster and engages in combat.
    - **If he rolls the die and gets an even number**, he activates his communicator to create a diversion and escapes.

- **If the die shows 2 or 4**, he takes the central corridor and discovers an abandoned control room.
    - **If he rolls the die and gets an even number**, he downloads the secret plans but sets off a silent alarm.
    - **If he rolls the die and gets an odd number**, he only takes some strategic information and continues his mission discreetly.

- **If the die roll shows a 5-6**, he takes the right corridor and comes across an Imperial droid blocking his path.
    - **If he rolls the die and gets an even number**, he manages to reprogram the droid that helps him access the evacuation hangar.
    - **If he rolls the die and gets an odd number**, the droid alerts the Imperial forces and the Rebel must return to the starting point.

This simple structure in the Star Wars universe perfectly illustrates the concept of conditional branching. Participants can easily project themselves into this familiar world and understand how choices and chance determine the course of the adventure. To make this activity even more engaging, you can:
Use typical Star Wars sounds (blaster sounds, Darth Vader breathing) during key moments
Present the different options with the characteristic intonations of the characters in the saga
Offer simple props like a cardboard lightsaber for the person narrating the story
Project the Empire or Rebel Alliance emblem in the background during the activity

Bluey's Adventure in the Park

Initial situation: Bluey and Bingo are playing an adventure game at the park and discover three different paths to explore.

- **If the die shows 1 or 3**, they take the left path and meet their friend Chloe who is organizing a treasure hunt.
  - **If they roll the die and get an even number**, Bluey and Bingo decide to take part in the treasure hunt and find a beautiful seashell.
  - **If they roll the die and get an odd number**, they suggest Chloe play "Magic Coffee" and create an imaginary potion that makes them grow taller instantly.

- **If the die shows 2 or 4**, they take the middle path and discover a large tree to climb.
  - **If they roll the die and get an even number**, Bluey helps Bingo climb, but a branch snaps under their weight. They must call on Dad (Bandit) to help them down.
  - **If they roll the die and get an odd number**, they manage to climb to the top and see the ice cream vendor in the distance. Off to the ice cream vendor!

- **If the die shows 5-6**, they take the right path and come across their neighbor Pat (Lucky's dad) who is walking his dog.
  - **If they roll the die and get an even number**, Pat suggests they play ball and they have a great time playing "Keepie Uppie."
  - **If they roll the die and get an odd number**, the dog escapes and they set off in a mad chase through the park.

This narrative maze set in the colorful and joyful world of Bluey allows participants to familiarize themselves with the concepts of conditional choice in a fun and supportive setting. The playful and creative spirit of the series is perfectly suited to introducing algorithmic thinking to teachers, who can easily transfer this approach to their students. To enhance this activity:

Use illustrations of Bluey characters to represent each path

Imitate the characteristic voices of Bluey and Bingo to tell the story

Include some typical phrases from the series like "How dare you!", "For real life?!" or "That's difficult!"

Invite participants to briefly play "Musical Statue" between two choices to stay in the atmosphere of the games in the series.

The Hero's Journey: A Universal Framework

The "hero's journey" or "monomyth," theorized by Joseph Campbell in "The Hero with a Thousand Faces," provides a powerful narrative structure for your storytelling labyrinth. This universal structure is found in many folktales, myths, and stories across cultures.

Key stages of the hero's journey:

- **The Ordinary World** - The starting point, the hero's familiar environment
- **The Call to Adventure** - A problem or challenge that compels the hero to action
- **Refusal of the call** - Initial hesitation (may be a decision point)
- **Meeting the Mentor** - Acquiring Knowledge or Magical Items
- **Crossing the First Threshold** - Entering the Extraordinary or Unknown World
- **Trials, Allies and Enemies** - A series of challenges that test the hero
- **Approaching the Heart of the Cave** - Preparing for the Major Test
- **The Supreme Test** - The Greatest Challenge to Overcome
- **The Reward** - What the hero gains after overcoming the ordeal
- **The Way Back** - The return journey, often with new obstacles
- **The Resurrection** - A final test that proves the hero's transformation
- **Returning with the Elixir** - Returning to the Ordinary World with a Treasure, a Lesson, or a Wisdom

Application to the narrative labyrinth:

Each step can become a potential decision point in your narrative maze. For example:

- At the "**call to adventure**", the hero can accept or refuse (first condition)
- During the "**meet the mentor**" different mentors may offer different tools (branching)
- During "**the tests**", several approaches can be attempted (multiple conditions)
- At the "**supreme test**", various strategies can be employed (major branch)

Other adaptable narrative structures:

- **Three-act structure** (exposition, confrontation, resolution)
- **Greimas' actantial schema** (subject, object, sender, recipient, adjuvant, opponent)
- **Propp's Functions** (31 Narrative Functions of Fairy Tales)

Create a storytelling - Character sheet

Initial situation: Once upon a time...

	Meet the Hero	Meet the Villain
Character name		
What motivates him? What are its values?		
What are its specialties, its qualities? What are its capabilities?		
What are its vulnerabilities?		
What is his previous background?		

Create a storytelling - The hero's journey

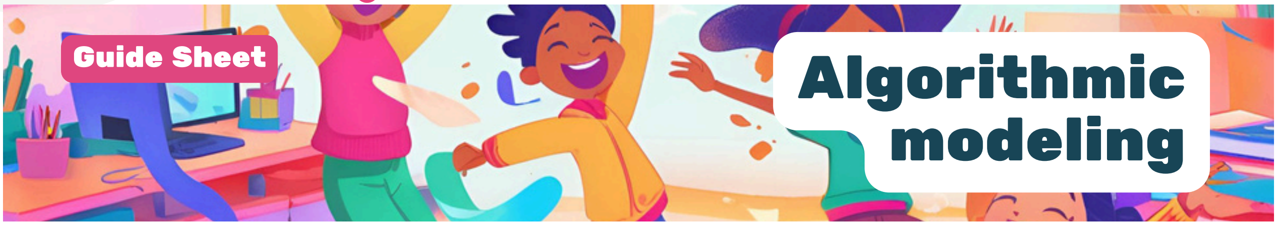
We meet our hero and discover his call to adventure...

Our hero meets his mentors...

And then the journey and the adventure begins...

...until the final confrontation with the villain...

...our hero returns to his life... However, changed forever!



A **flowchart** is a visual representation of an algorithm that shows the **sequence of operations and decisions** using **standardized symbols**. For the narrative maze, the flowchart visualizes **all possible narrative paths and their interconnections.**

Standard symbols of a flowchart

Symbol	Shape	Signification
Start/End	Oval or round	Marks the beginning or end of the process
Treatment	Rectangle	Represents an action or narrative step
Decision	Diamond	A choice point with several possible outcomes (conditional branching)
Connector	Circle	A junction point between different parts of the diagram
Flux	Arrow	Indicates the direction of flow between steps

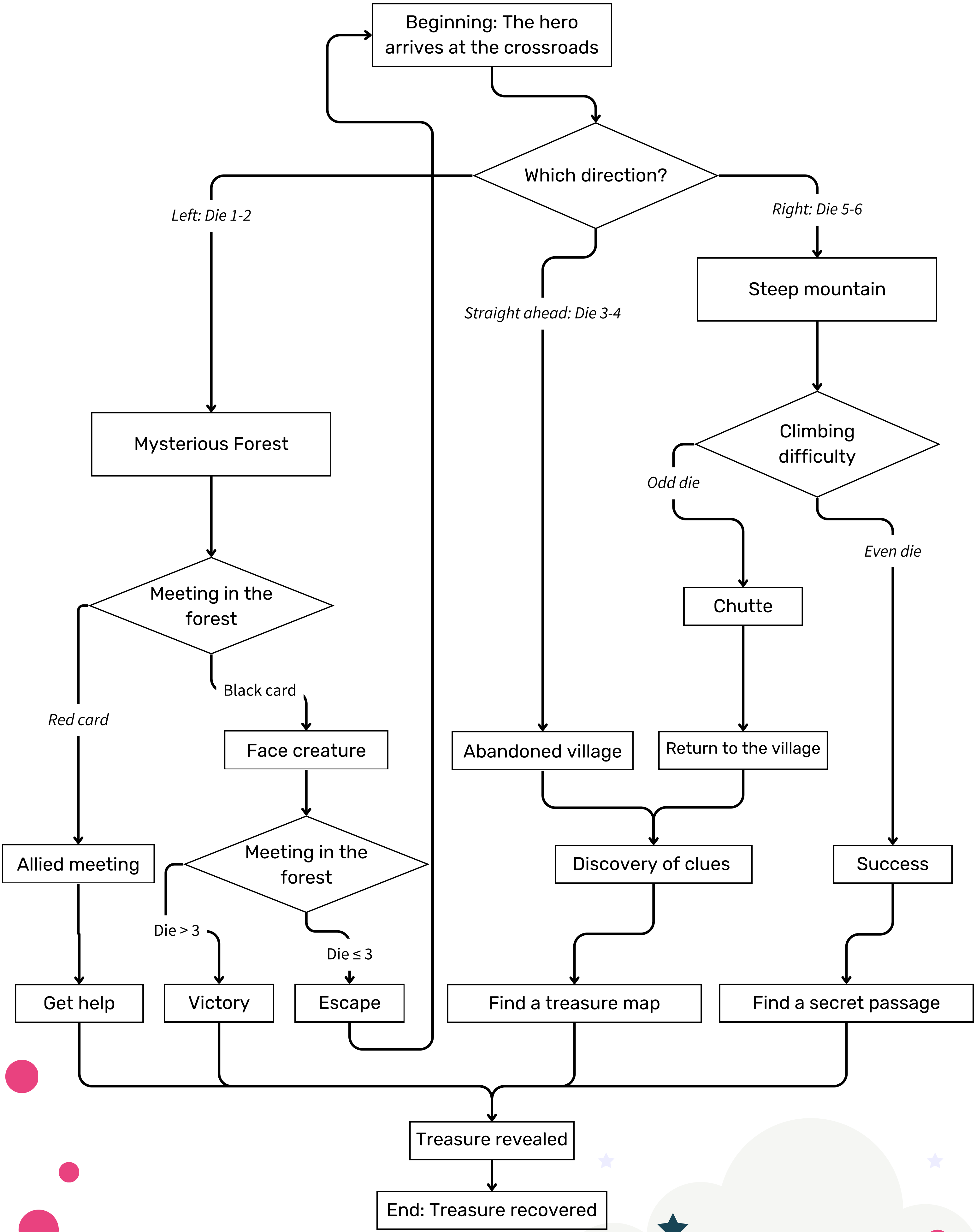
Creating a flowchart with Mermaid (for teachers)

Mermaid is a simple language for creating diagrams in a digital environment. It can be used in many tools (GitHub, Notion, VSCode, etc.) and allows you to quickly generate flowcharts. The basic syntax in Mermaid is as follows:

```
flowchart TD
    A[Start] --> B{First choice}
    B -->|Option 1| C[Action 1]
    B -->|Option 2| D[Action 2]
    C --> E{Second choice}
    D --> F[Consequence]
    E -->|Option A| G[Good end]
    E -->|Option B| H[Bad end]
    F --> G
```

flowchart TD: Defines a top-down flowchart
A[Start]: Creates a rectangular node named "A" containing the text "Start"
A --> B: Creates an arrow from node A to node B
B{First choice}: Creates a diamond-shaped node (decision)
B -->|Option 1| C: Arrow from B to C with the label "Option 1"

Example of a flowchart for a simple narrative maze



Practical application for the activity

- **Make a sketch**: Start by drawing a rough draft on paper to visualize the overall structure.

- **Clearly identify:**
  - The starting point (beginning of the story)
  - Decision points (with their conditions)
  - The resulting actions
  - Narrative convergences
  - The possible ends

- **Create the digital diagram:**
  - Using Mermaid (online via mermaid.live or in compatible tools)
  - Code each element with the appropriate syntax
  - Check the consistency of the flows

- **Analyze the diagram:**
  - Identify dead branches or narrative dead ends
  - Identify paths that are too short or too long
  - Balancing the different branches in terms of complexity and interest

- **Iterate and improve:**
  - Adjust problematic branches
  - Add connections between branches if needed
  - Simplify the overly complex parts

This modeling process not only allows us to visualize the narrative labyrinth but also to concretely introduce the concepts of conditional structures, execution flow and modularity which are at the heart of algorithmic thinking.